

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING**  
**BACHELOR OF SCIENCE IN CYBER SECURITY**  
**MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO**  
**NETWORK SECURITY (CYS360)**  
**(6<sup>TH</sup> SEMESTER, 3<sup>RD</sup> Year) LAB EXPERIMENT # 13**

|        |                             |       |                   |                         |                                           |
|--------|-----------------------------|-------|-------------------|-------------------------|-------------------------------------------|
| Name:  | <u>  _Muhammad_Usman_  </u> |       | Roll number:      | <u>  _23BSCYS006_  </u> |                                           |
| Score: | <u>          </u>           | Date: | <u>          </u> | Instructor's Signature: | <u>                                  </u> |

## Lab # 13: To configure a Point-to-Point IPSec Tunnel Manually

**Objective:** The objective of this lab is to configure Point-to-Point IPSec Tunnel Manually

**Duration: 3 Hours**

**Expected Outcomes:** By the end of this lab session, students will be able to:

1. Students will be able to manually configure an IPSec VPN tunnel between two network device.
2. Students will be able to verify and troubleshoot IPSec tunnel connectivity

### Rubrics

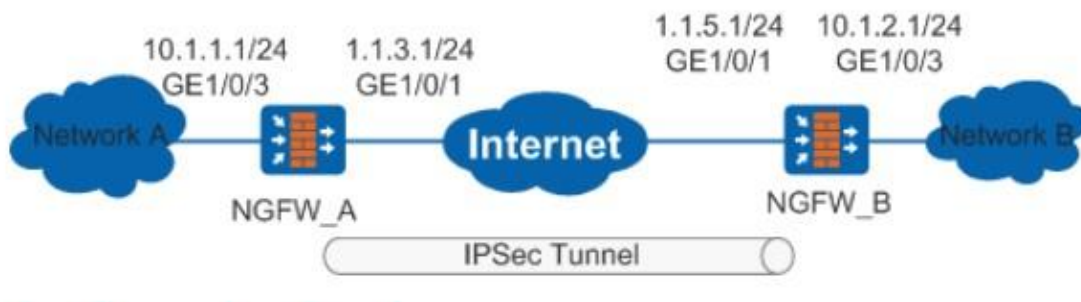
| LAB PERFORMANCE INDICATOR | SUBJECT KNOWLEDGE         | DATA ANALYSIS AND INTERPRETATION | ABILITY TO CONDUCT EXPERIMENT |
|---------------------------|---------------------------|----------------------------------|-------------------------------|
| SCORE                     | <u>                  </u> | <u>                  </u>        | <u>                  </u>     |

*Note: Student performance will be assessed based on this rubric during lab sessions.*

### Networking Requirements

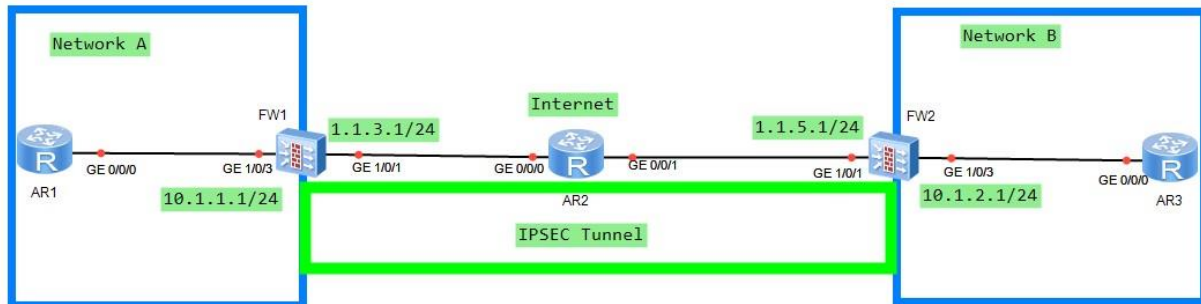
As shown in [Figure 1](#), network A and network B connect to the Internet through FW\_A and FW\_B respectively. The network environment is as follows:

- Network A (10.1.1.0/24) connects to FW\_A through GE1/0/3.
- Network B (10.1.2.0/24) connects to FW\_B through GE1/0/3.
- FW\_A and FW\_B are reachable to each other.



*Figure 1 Networking diagram for configuring IPSEC Tunnel*

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*Figure 2 Networking Topology of IPSEC Tunnel*

## Configuration Roadmap

The configuration roadmap is as follows:

1. According to network planning, allocate interfaces to the FW, configure IP addresses for the interfaces, and add the interfaces to corresponding security zones.
2. Configure an ACL to define data flows that need to be protected.
3. Configure interzone security policies.
4. Configure IPsec proposals.
5. Configure IPsec policies.
6. Apply IPsec policies.

## Data Plan

To complete the configuration, you need the following data:

- **IP addresses of interfaces on the FW**

[Figure 1](#) shows the interfaces and IP addresses of this example.

- **Security protocols, and encryption and authentication algorithms adopted by security protocols**

In this example, ESP is adopted as the security protocol. The encryption algorithm and authentication algorithm adopted by ESP are AES-256 and SHA2-256 respectively.

- **Inbound and outbound Security Parameter Indexes (SPIs)**

In this example, the SPIs of the inbound and outbound SAs adopted by FW\_A are **12345678** and **87654321** respectively. The SPIs of the inbound and outbound SAs adopted by FW\_B are **87654321** and **12345678** respectively.

- **ESP authentication key**

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In this example, the inbound and outbound authentication keys of ESP adopted by FW\_A are **abcdefg** and **gfedcba** respectively. The inbound and outbound authentication keys of ESP adopted by FW\_B are **gfedcba** and **abcdefg** respectively.

## Procedure

- Configure FW\_A.

1. Configure the IP address.

- a. Configure the IP address of GE1/0/3.

```
<sysname> system-view
[sysname] sysname FW_A
[FW_A] interface gigabitethernet 1/0/3
[FW_A-GigabitEthernet1/0/3] ip address 10.1.1.1 24
[FW_A-GigabitEthernet1/0/3] quit
```

- b. Configure the IP address of GE1/0/1.

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ip address 1.1.3.1 24 [FW_A-
GigabitEthernet1/0/1] quit
```

2. Add interfaces to corresponding security zones.

Add GE1/0/3 to the Trust zone.

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface gigabitethernet 1/0/3
[FW_A-zone-trust] quit
```

Add GE1/0/1 to the Untrust zone.

```
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface gigabitethernet 1/0/1 [FW_A-zone-
untrust] quit
```

3. Configure interzone security policies.

- a. Configure the security policies between the Trust and Untrust zones.

```
[FW_A] security-policy
[FW_A-policy-security] rule name policy1
[FW_A-policy-security-rule-policy1] source-zone trust
[FW_A-policy-security-rule-policy1] destination-zone untrust
```

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```
[FW_A-policy-security-rule-policy1] source-address 10.1.1.0 24
[FW_A-policy-security-rule-policy1] destination-address 10.1.2.0 24 [FW_A-
policy-security-rule-policy1] action permit
[FW_A-policy-security-rule-policy1] quit
[FW_A-policy-security] rule name policy2
[FW_A-policy-security-rule-policy2] source-zone untrust
[FW_A-policy-security-rule-policy2] destination-zone trust
[FW_A-policy-security-rule-policy2] source-address 10.1.2.0 24
[FW_A-policy-security-rule-policy2] destination-address 10.1.1.0 24
[FW_A-policy-security-rule-policy2] action permit
[FW_A-policy-security-rule-policy2] quit
```

**b. Configure the security policies between the Local and Untrust zones.**

```
FW_A-policy-security] rule name policy3
[FW_A-policy-security-rule-policy3] source-zone local
[FW_A-policy-security-rule-policy3] destination-zone untrust
[FW_A-policy-security-rule-policy3] source-address 1.1.3.1 32
[FW_A-policy-security-rule-policy3] destination-address 1.1.5.1 32
[FW_A-policy-security-rule-policy3] action permit
[FW_A-policy-security-rule-policy3] quit
[FW_A-policy-security] rule name policy4
[FW_A-policy-security-rule-policy4] source-zone untrust
[FW_A-policy-security-rule-policy4] destination-zone local
[FW_A-policy-security-rule-policy4] source-address 1.1.5.1 32
[FW_A-policy-security-rule-policy4] destination-address 1.1.3.1 32
[FW_A-policy-security-rule-policy4] action permit
[FW_A-policy-security-rule-policy4] quit
[FW_A-policy-security] quit
```

**4. Configure a static route to network B. Assume that the next hop of the route is 1.1.3.2.**

```
[FW_A] ip route-static 10.1.2.0 255.255.255.0 1.1.3.2
[FW_A] ip route-static 1.1.5.0 255.255.255.0 1.1.3.2
```

**5. Configure advanced ACL 3000 to permit the users on network segment 10.1.1.0/24 to access network segment 10.1.2.0/24.**

```
[FW_A] acl 3000
[FW_A-acl-adv-3000] rule 5 permit ip source 10.1.1.0 0.0.0.255 destination
10.1.2.0 0.0.0.255
[FW_A-acl-adv-3000] quit
```

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**6. Configure IPSec proposal **tran1**.**

```
[FW_A] ipsec proposal tran1
[FW_A-ipsec-proposal-tran1] encapsulation-mode tunnel
[FW_A-ipsec-proposal-tran1] transform esp
[FW_A-ipsec-proposal-tran1] esp authentication-algorithm sha2-256
[FW_A-ipsec-proposal-tran1] esp encryption-algorithm aes-256
[FW_A-ipsec-proposal-tran1] quit
```

**7. Configure an IPSec policy with the name **map1** and number **10**.**

```
[FW_A] ipsec policy map1 10 manual
[FW_A-ipsec-policy-manual-map1-10] security acl 3000
[FW_A-ipsec-policy-manual-map1-10] proposal tran1
[FW_A-ipsec-policy-manual-map1-10] tunnel remote 1.1.5.1
[FW_A-ipsec-policy-manual-map1-10] tunnel local 1.1.3.1
[FW_A-ipsec-policy-manual-map1-10] sa spi inbound esp 12345678
[FW_A-ipsec-policy-manual-map1-10] sa spi outbound esp 87654321
[FW_A-ipsec-policy-manual-map1-10] sa string-key inbound esp abcdefg
[FW_A-ipsec-policy-manual-map1-10] sa string-key outbound esp gfedcba
[FW_A-ipsec-policy-manual-map1-10] quit
```

**8. Apply security policy **map1** to GE1/0/1.**

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ipsec policy map1
[FW_A-GigabitEthernet1/0/1] quit
```

**Configure FW\_B.**

**1. Set interface IP addresses.**

**a. Configure the IP address of GE1/0/3.**

```
<sysname> system-view
[sysname] sysname FW_B
[FW_B] interface gigabitethernet 1/0/3
[FW_B-GigabitEthernet1/0/3] ip address 10.1.2.1 24
[FW_B-GigabitEthernet1/0/3] quit
```

**b. Configure the IP address of GE1/0/1.**

```
[FW_B] interface gigabitethernet 1/0/1
[FW_B-GigabitEthernet1/0/1] ip address 1.1.5.1 24
[FW_B-GigabitEthernet1/0/1] quit
```

**2. Add interfaces to corresponding security zones.**

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- a. Add GE1/0/3 to the Trust zone.

```
[FW_B] firewall zone trust
[FW_B-zone-trust] add interface gigabitethernet 1/0/3
[FW_B-zone-trust] quit
```

- b. Add GE1/0/1 to the Untrust zone.

```
[FW_B] firewall zone untrust
[FW_B-zone-untrust] add interface gigabitethernet 1/0/1
[FW_B-zone-untrust] quit
```

3. Configure interzone security policies.

- a. Configure the security policies between the Trust and Untrust zones.

```
[FW_B] security-policy
[FW_B-policy-security] rule name policy1
[FW_B-policy-security-rule-policy1] source-zone trust
[FW_B-policy-security-rule-policy1] destination-zone untrust
[FW_B-policy-security-rule-policy1] source-address 10.1.2.0 24
[FW_B-policy-security-rule-policy1] destination-address 10.1.1.0 24
[FW_B-policy-security-rule-policy1] action permit
[FW_B-policy-security-rule-policy1] quit
[FW_B-policy-security] rule name policy2
[FW_B-policy-security-rule-policy2] source-zone untrust
[FW_B-policy-security-rule-policy2] destination-zone trust
[FW_B-policy-security-rule-policy2] source-address 10.1.1.0 24
[FW_B-policy-security-rule-policy2] destination-address 10.1.2.0 24
[FW_B-policy-security-rule-policy2] action permit
[FW_B-policy-security-rule-policy2] quit
```

- b. Configure the security policies between the Local and Untrust zones.

```
[FW_B-policy-security] rule name policy3
[FW_B-policy-security-rule-policy3] source-zone local
[FW_B-policy-security-rule-policy3] destination-zone untrust
[FW_B-policy-security-rule-policy3] source-address 1.1.5.1 32
[FW_B-policy-security-rule-policy3] destination-address 1.1.3.1 32
[FW_B-policy-security-rule-policy3] action permit
[FW_B-policy-security-rule-policy3] quit
[FW_B-policy-security] rule name policy4
[FW_B-policy-security-rule-policy4] source-zone untrust
[FW_B-policy-security-rule-policy4] destination-zone local
[FW_B-policy-security-rule-policy4] source-address 1.1.3.1 32
[FW_B-policy-security-rule-policy4] destination-address 1.1.5.1 32
[FW_B-policy-security-rule-policy4] action permit
```

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```
[FW_B-policy-security-rule-policy4] quit  
[FW_B-policy-security] quit
```

4. Configure a static route to network A. Assume that the next hop of the route is 1.1.5.2.

```
[FW_B] ip route-static 10.1.1.0 255.255.255.0 1.1.5.2  
[FW_B] ip route-static 1.1.3.0 255.255.255.0 1.1.5.2
```

5. Configure advanced ACL 3000 to permit the users on network segment 10.1.2.0/24 to access network segment 10.1.1.0/24.

```
[FW_B] acl 3000  
[FW_B-acl-adv-3000] rule 5 permit ip source 10.1.2.0 0.0.0.255 destination  
10.1.1.0 0.0.0.255  
[FW_B-acl-adv-3000] quit
```

6. Configure IPSec proposal **tran1**.

```
[FW_B] ipsec proposal tran1  
[FW_B-ipsec-proposal-tran1] encapsulation-mode tunnel  
[FW_B-ipsec-proposal-tran1] transform esp  
[FW_B-ipsec-proposal-tran1] esp authentication-algorithm sha2-256  
[FW_B-ipsec-proposal-tran1] esp encryption-algorithm aes-256  
[FW_B-ipsec-proposal-tran1] quit
```

7. Configure an IPSec policy with the name **map1** and number **10**.

```
[FW_B] ipsec policy map1 10 manual  
[FW_B-ipsec-policy-manual-map1-10] security acl 3000  
[FW_B-ipsec-policy-manual-map1-10] proposal tran1  
[FW_B-ipsec-policy-manual-map1-10] tunnel remote 1.1.3.1  
[FW_B-ipsec-policy-manual-map1-10] tunnel local 1.1.5.1  
[FW_B-ipsec-policy-manual-map1-10] sa spi inbound esp 87654321  
[FW_B-ipsec-policy-manual-map1-10] sa spi outbound esp 12345678  
[FW_B-ipsec-policy-manual-map1-10] sa string-key inbound esp gfedcba  
[FW_B-ipsec-policy-manual-map1-10] sa string-key outbound esp abcdefg  
[FW_B-ipsec-policy-manual-map1-10] quit
```

8. Apply IPSec policy **map1** to GE1/0/1.

```
[FW_B] interface gigabitethernet 1/0/1  
[FW_B-GigabitEthernet1/0/1] ipsec policy map1  
[FW_B-GigabitEthernet1/0/1] quit
```

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## Verification

- On FW\_A and FW\_B, run the **display ipsec sa** command to check whether SAs are established. Take FW\_B as an example. If the following information is displayed, the IKE SA and IPSec SA are successfully established.

```
<FW_B> display ipsec sa

                                     ipsec sa information:
=====
Interface: GigabitEthernet1/0/1
=====

-----
IPSec policy name: "map1"
Sequence number   : 10
Acl group         : 3000
Acl rule          : -
Mode              : Manual
-----

Encapsulation mode: Tunnel
Tunnel local       : 1.1.5.1
Tunnel remote      : 1.1.3.1

[Outbound ESP SAs]
SPI: 12345678 (0xbc614e)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA encrypted packets (number/bytes): 0/0
No duration limit for this SA

[Inbound ESP SAs]
SPI: 87654321 (0x5397fb1)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA decrypted packets (number/bytes): 0/0
No duration limit for this SA
Anti-replay : Disable
```

- If the configurations are correct, network A can ping network B, and you can check the encrypted data using the **display ipsec statistics** command.

```
<FW_B> display ipsec statistics
```



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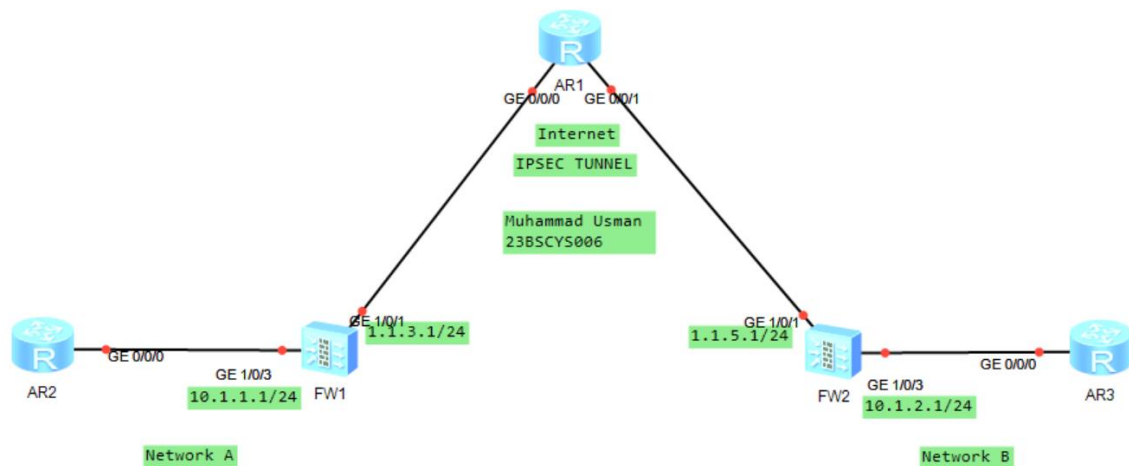
```
IPSec statistics information:
the security packet statistics:
  input/output security packets: 20/20
input/output security bytes: 2000/2000
input/output dropped security packets: 0/0
encrypt packet statistics:
  send chip: 63, recv chip: 63, send err: 0
local cpu: 63, other cpu: 0, recv other cpu: 0
intact packet: 63, first slice: 0, after slice: 0
the decrypt packet statistics:
  send chip: 244, recv chip: 244, send err: 0
cpu: 244, other cpu: 0, recv other cpu: 0
first slice: 0, after slice: 0
packet detail:
SA: 0, wrong SA: 0
0, replay: 0
recheck: 0, after recheck: 0
change cpu enc: 0, dec change cpu: 0
fib search: 0, output l3: 0
flow err: 0, slice err: 0, byte limit: 0
negotiate about packet statistics:
  IKE fwd packet ok: 93, err: 0
  IKE ctrl packet inbound ok: 92, outbound ok: 58
SoftExpr: 0, HardExpr: 0, DPDOper: 0
trigger ok: 0, switch sa: 35684, sync sa: 0
  recv IKE nat keepalive: 114731, IKE input: 0
```

**Task:**

Attach Screenshots of all the topology and configuration.

**Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.**

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```
[23BSCYS006-FW-A]dis ipsec sa
2026-04-11 15:10:38.500

ipsec sa information:

=====
Interface: GigabitEthernet1/0/1
=====

-----
IPSec policy name: "map1"
Sequence number   : 10
Acl group         : 3000
Acl rule          : -
Mode              : Manual
-----

Encapsulation mode: Tunnel
Tunnel local       : 1.1.3.1:0
Tunnel remote      : 1.1.5.1:0

[Outbound ESP SAs]
SPI: 87654321 (0x5397fb1)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA encrypted packets (number/bytes): 0/0
No duration limit for this SA

[Inbound ESP SAs]
SPI: 12345678 (0xbc614e)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA decrypted packets (number/bytes): 0/0
No duration limit for this SA
Anti-replay : Disable
[23BSCYS006-FW-A]
```

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```
[23BSCYS006-FW-A]dis ipsec statistics
2026-04-11 15:11:18.300
IPSec statistics information:
Number of IPSec tunnels: 1
Number of standby IPSec tunnels: 0
the security packet statistics:
  input/output security packets: 0/0
  input/output security bytes: 0/0
  input/output dropped security packets: 0/0
the encrypt packet statistics:
  send chip: 0, rcv chip: 0, send err: 0
  local cpu: 0, other cpu: 0, rcv other cpu: 0
  intact packet: 0, first slice: 0, after slice: 0
the decrypt packet statistics:
  send chip: 0, rcv chip: 0, send err: 0
  local cpu: 0, other cpu: 0, rcv other cpu: 0
  reass first slice: 0, after slice: 0
dropped security packet detail:
  can not find SA: 0, wrong SA: 0
  authentication: 0, replay: 0
  front recheck: 0, after recheck: 0
  change cpu enc: 0, dec change cpu: 0
  fib search: 0, output 13: 0
  flow err: 0, slice err: 0, byte limit: 0
  slave drop: 0
negotiate about packet statistics:
  IKE fwd packet ok: 0, err: 0
  IKE ctrl packet inbound ok: 0, outbound ok: 0
  SoftExpr: 0, HardExpr: 0, DPDOper: 0
  trigger ok: 0, switch sa: 0, sync sa: 0
  rcv IKE nat keepalive: 0, IKE input: 0
```

[23BSCYS006-FW-A]

```
[23BSCYS006-FW-B]dis ipsec sa
2026-04-11 15:12:20.670

ipsec sa information:

=====
Interface: GigabitEthernet1/0/1
=====

-----
IPSec policy name: "map1"
Sequence number   : 10
Acl group         : 3000
Acl rule          : -
Mode              : Manual
-----

Encapsulation mode: Tunnel
Tunnel local      : 1.1.5.1:0
Tunnel remote     : 1.1.3.1:0

[Outbound ESP SAs]
SPI: 12345678 (0xbc614e)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA encrypted packets (number/bytes): 0/0
No duration limit for this SA

[Inbound ESP SAs]
SPI: 87654321 (0x5397fb1)
Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
SA decrypted packets (number/bytes): 0/0
No duration limit for this SA
Anti-replay : Disable
[23BSCYS006-FW-B]
```

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```
[23BSCYS006-FW-B]dis ipsec statistics
2026-04-11 15:12:58.130
IPSec statistics information:
Number of IPSec tunnels: 1
Number of standby IPSec tunnels: 0
the security packet statistics:
  input/output security packets: 0/0
  input/output security bytes: 0/0
  input/output dropped security packets: 0/0
the encrypt packet statistics:
  send chip: 0, recv chip: 0, send err: 0
  local cpu: 0, other cpu: 0, recv other cpu: 0
  intact packet: 0, first slice: 0, after slice: 0
the decrypt packet statistics:
  send chip: 0, recv chip: 0, send err: 0
  local cpu: 0, other cpu: 0, recv other cpu: 0
  reass first slice: 0, after slice: 0
dropped security packet detail:
  can not find SA: 0, wrong SA: 0
  authentication: 0, replay: 0
  front recheck: 0, after recheck: 0
  change cpu enc: 0, dec change cpu: 0
  fib search: 0, output l3: 0
  flow err: 0, slice err: 0, byte limit: 0
  slave drop: 0
negotiate about packet statistics:
  IKE fwd packet ok: 0, err: 0
  IKE ctrl packet inbound ok: 0, outbound ok: 0
  SoftExpr: 0, HardExpr: 0, DPDOper: 0
  trigger ok: 0, switch sa: 0, sync sa: 0
  recv IKE nat keepalive: 0, IKE input: 0

[23BSCYS006-FW-B]
```